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Screening for Lung Cancer, Overdiagnosis, and Healthcare Utilization: A Nationwide Population-Based Study

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Abstract

Introduction: Guideline-discordant low-dose computed tomography (LDCT) screening may cause lung cancer (LC) overdiagnosis, but its extent and consequences are unclear. This study aimed to investigate the prevalence of self-initiated, non-reimbursed LDCT screening in a predominantly non-smoking population and its impact on LC epidemiology and healthcare utilization.

Methods: This nationwide cohort study analyzed data from Korea's National Health Information Database (NHID) and eleven academic hospital screening centers (1999–2022). The overall analysis encompassed the entire Korean population. For non-reimbursed LDCT screening prevalence, which NHID does not capture, a separate analysis was conducted on a cohort of 1.7 million adults to extrapolate nationwide rates. Outcomes included trends in self-initiated, non-reimbursed LDCT screening, LC incidence, mortality, stage and age at diagnosis, 5-year survival, and LC-related healthcare utilization, including surgeries and biopsies. Joinpoint regression assessed trend changes.

Results: Self-initiated, non-reimbursed LDCT screening during health check-ups increased from 29% to 60% in men and 7% to 46% in women, despite only 2.4% of men and 0.04% of women qualifying for risk-based screening. In women, localized-stage LC incidence nearly doubled (age-standardized incidence rate [ASIR], from 7.6 to 13.7 per 100,000), while distant-stage incidence decreased (ASIR, from 16.1 to 15.0 per 100,000). LC mortality declined (age-standardized mortality rate, from 23.3 to 19.8 per 100,000), while 5-year survival rates improved significantly. LC diagnoses in women shifted towards earlier stages and younger ages. Lung surgeries for both malignant and benign lesions, frequently lacking nonsurgical biopsies, increased sharply in women.

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Conclusions: Widespread guideline-discordant LDCT screening correlates with LC overdiagnosis and increased healthcare utilization, particularly in women. Randomized controlled trials are needed to assess the risks and benefits of screening in low-risk populations to determine its efficacy and consequences.

Keywords

lung cancer screening; overdiagnosis; healthcare utilization

Introduction

Globally, lung cancer (LC) ranks first in incidence among men and third in women, representing 14.3% and 8.4% of annual cancer diagnoses, respectively.[1] Despite its heterogeneity and diverse progression patterns, LC remains the leading cause of cancer-related deaths, with survival dependent on the stage at diagnosis.[2,3] This understanding raised anticipation that early detection through screening could lower mortality.[4] Screening trials using low-dose computed tomography (LDCT) demonstrated LC mortality reduction in high-risk groups,[5–8] prompting its global implementation and shifting the priority toward identifying those who would benefit most.[9]

In Asia, however, widespread LDCT screening, often deviating from risk-based guidelines, raised concerns about LC overdiagnosis.[10–13] Asia's recent LC "epidemic" is characterized by: 1) heightened demand and supply for guideline-discordant LDCT screening,[10–12] 2) prevalent incidental detection of previously undetectable, indolent early adenocarcinoma spectrum lesions presenting as pulmonary subsolid nodules (SSNs), where long-term survival rates near 100% even in confirmed malignancy cases,[14–19] and 3) disproportionately rising early-stage LC in women, despite stable advanced-stage diseases and mortality.[10–12,20] This trend deviates from typical cancer patterns, where incidence and mortality gradually change in parallel with risk alterations.[21,22] Instead, it primarily affects women with minimal smoking exposure, coinciding with growing public screening interest fueled by: 1) comparable LC detection rates in populations not recommended for screening, 2) increased awareness of rising LC incidence in low-risk groups without clear etiologic explanations, and 3) aggressive promotion of affordable LDCT by healthcare centers since the 2000s.[23–26] The resulting epidemiologic shift in low-risk populations, showing divergent stage-specific incidence with stable mortality, indicates overdiagnosis.[21,22]

Overdiagnosis—detecting disease not destined to become clinically evident—occurs when pervasive screening identifies a large reservoir of slow-growing cancers, increasing early-stage incidence without affecting mortality.[27,28] Addressing overdiagnosis is crucial to avoid unnecessary invasive interventions, associated harms, and psychosocial distress.[29] Despite accumulating evidence, research struggled to quantify excessive screening rates, which is essential to confirm overdiagnosis. Furthermore, no studies have addressed the impact of screening low-risk populations on healthcare utilization. These challenges stem from difficulties in obtaining large-scale screening and healthcare usage data.

Korea's National Health Information Database (NHID), a centrally organized, comprehensive healthcare database covering the entire population, provides a unique

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opportunity to address these issues. Integrating NHID with non-reimbursed screening data could uncover the extent of excessive screening and its implications. This nationwide study aimed to investigate the prevalence of self-initiated, non-reimbursed LDCT screening in a predominantly non-smoking population, and examine its impact on LC incidence, mortality, stage and age at diagnosis, 5-year survival, and subsequent invasive diagnostic and therapeutic procedures.

Material and Methods

Study Design

This nationwide, population-based ecological cohort study investigated LC screening, epidemiology, and healthcare utilization in Korea. First, we evaluated national smoking prevalence, risk-based screening eligibility rates, and self-initiated, non-reimbursed LDCT prevalence during regular health check-ups. Second, we analyzed trends in LC incidence, mortality, stage, and age at diagnosis, with 5-year survival, stratified by stage. Third, we tracked LC-related healthcare utilization, specifically invasive diagnostic and therapeutic procedures, with shifts in the type of surgery performed. The study period, 1999–2022, spanned before and after widespread LDCT screenings. This study adhered to Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines and received IRB approval from Seoul National University Hospital (2308-110-1459) and participating centers.[30]

Data Sources

Four primary data sources were used: 1) the Korean Health Insurance Review and Assessment (HIRA) database, 2) Statistics Korea, 3) the Korea National Cancer Incidence Database (KNCI DB), and 4) health check-up records from eleven nationwide academic hospital screening centers. Korea's National Health Insurance Service (NHIS) operates as a single centralized insurer, providing universal healthcare coverage and tracking healthcare utilization for the entire population through unique resident registration numbers.[31] It offers free biennial health check-ups (75% participation rate over a decade) including cancer screenings. Korea's national LC screening program, initiated in 2019, recommends annual LDCT for adults aged 54–74 years who have smoked over 30 pack-years, including ex-smokers who quit less than 15 years ago (52.6% participation rate) (Supplementary Text 1).[32]

HIRA captures claim records for 98% of the population under NHIS, including diagnostic codes and data on healthcare usage, such as prescriptions, procedures, and surgeries.[33] These integrate into the national database, constituting a comprehensive healthcare data infrastructure, with health statistics coming from Statistics Korea and cancer-specific data from the Korean Central Cancer Registry (KCCR) and KNCI DB. KCCR, Asia's largest national cancer registry, collects data on cancer incidence, prevalence, and survival.[34] From 1999 onward, KNCI DB documents detailed data on incident cancer, including age and stage at diagnosis, and histologic subtype, with a completeness rate of 98.3%, forming the basis for annual national cancer reports. While the overall analysis included the entire population, data on non-reimbursed LDCT were not captured in NHID and

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were derived from a multi-institutional cohort. Building upon Korea's healthcare system, most academic hospitals operate screening centers—with referral systems for managing detected abnormalities—offering both reimbursed and non-reimbursed services. During health check-ups, eligible individuals receive reimbursed LDCT, while low-risk individuals can undergo non-reimbursed LDCT for approximately $74. To capture real-world screening practices, health check-up records were collected from eleven representative academic hospital screening centers across four of Korea's five primary administrative districts.

Outcomes

Smoking prevalence and LC screening: National smoking prevalence and risk-based screening eligibility rates were obtained from Statistics Korea. Self-initiated, non-reimbursed LDCT screening prevalence was evaluated among adults aged ≥19 years who underwent regular health check-ups at study centers between 2004 and 2020. Annual figures for health check-ups and LDCT recipients were compiled.

LC epidemiology: LC incidence, mortality, and 5-year survival rates from 1999 to 2019 were secured from Statistics Korea, with stage and age at diagnosis from KNCI DB since 2005. LC cases were identified using the International Classification of Diseases, 10th edition, codes C33 and C34. Stages were classified following the Surveillance, Epidemiology, and End Results (SEER) summary stage: 1) localized: confined to the organ of origin, stages 0 and I, 2) regional: extending beyond or near the organ of origin, stages II and III, and 3) distant: metastasis to non-adjacent organs, stage IV.[35] To mitigate bias resulting from missing stages, adjustments were made by allocating missing stages based on the annual stage-specific counts and stage distributions.[36] Stage-specific incidence was determined by multiplying the overall incidence by the annual stage distribution. Survival rates were analyzed over three periods: baseline (2006–2010), the onset of widespread LDCT screening, subsequent (2011–2015), and recent (2016–2020).

LC-related healthcare utilization: Healthcare utilization data were available starting in 2010 from HIRA, using specific claim codes. Annual counts of each practice were collected and age-standardized. All forms of lung surgeries, including thoracotomy, video- and robot-assisted thoracic surgeries were compiled. Surgical techniques included wedge resection, segmentectomy, lobectomy, and pneumonectomy. LC surgeries were identified as mediastinal lymph node dissection with pulmonary resection.[37] Nonsurgical procedures captured included percutaneous lung biopsy, transbronchial lung biopsy, transbronchial needle aspiration, bronchoscopy, thoracoscopy, and mediastinoscopy.

Statistical Analysis

All population rates were age-adjusted to the 2020 Korean standard population and expressed per 100,000 individuals.[38] Age-standardized rates were calculated as weighted averages of age-specific rates, with weights representing the proportions of each age group within the standard population.[38,39] Joinpoint regression analyzed log-linear temporal trends and time points of significant change (Supplementary Text 2).[40] The strength of changes was assessed using annual percentage change (APC) and average annual percent change (AAPC), with 95% confidence intervals (CIs). Statistical significance of APC

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and AAPC was determined using the Monte Carlo Permutation method, with two-sided $p<0.05$ considered significant.[40,41] The terms “increase” and “decrease” were used only for significant APC or AAPC ($p<0.05$). Statistical analysis was conducted using R (v4.1.2, R Foundation) and the Joinpoint Regression Program (v5.2.0, National Cancer Institute).

Results

Nationwide smoking prevalence among men decreased from 61.3% in 2001 to 30.7% in 2021, while remaining around 6% for women (Supplementary Figure 1). In 2020, 2.4% (521,056 of 21,616,672) of Korean men and 0.04% (8,690 of 22,007,361) of women qualified for national LC screening. Among individuals aged 54–74, 7.9% of men and 0.1% of women were eligible.

From 2004 to 2020, 44% (398,772 of 906,960) of men and 24% (190,158 of 801,696) of women underwent non-reimbursed LDCT during health check-ups at study centers. Participation in health check-ups and LDCT increased over time (Figure 1). The proportion of individuals opting for LDCT increased from 29% to 60% in men (APC, 4.1%; 95%CI, 3.1–5.2) and from 7% to 46% in women (APC, 8.3%; 95%CI, 5.8–11.0), surpassing both national smoking rates and eligibility for risk-based screening.

Between 1999 and 2019, 305,438 men and 127,564 women in Korea were diagnosed with LC, leading to 233,367 and 83,854 LC-related deaths, respectively. LC incidence and mortality trends differed by gender (Figure 2). In men, both the age-standardized incidence rate (ASIR) and age-standardized mortality rate (ASMR) decreased, from 98.9 to 84.4 per 100,000 (APC, −0.8%; 95%CI, −0.9–−0.6) and from 81.0 to 56.4 per 100,000 (APC, −1.9%; 95%CI, −2.2–−1.6), respectively. Conversely, women’s ASIR increased from 29.8 to 39.3 per 100,000 (AAPC, 1.6%; 95%CI, 1.4–1.7), while ASMR decreased from 23.3 to 19.8 per 100,000 (AAPC, −1.0%; 95%CI, −1.2–−0.7).

Figure 3A presents changes in LC stage-specific ASIR from 2005 to 2019. In men, the ASIR decreased for all stages: localized from 23.5 to 20.5 (AAPC, −1.0%; 95%CI, −1.7–−0.4), regional from 33.3 to 24.0 (APC, −2.1%; 95%CI, −2.7–−1.6), and distant from 41.3 to 37.1 per 100,000 (AAPC, −0.7%; 95%CI, −1.0–−0.4). In women, localized-stage ASIR nearly doubled from 7.6 to 13.7 per 100,000 (APC, 4.0%; 95%CI, 3.0–5.0). Regional-stage ASIR remained stable from 8.8 to 9.3 per 100,000 women. Distant-stage ASIR decreased slightly from 16.1 to 15.0 per 100,000 (AAPC, −0.6%; 95%CI, −1.0–−0.1), with a modest increase noted until 2011 (APC, 1.5%; 95%CI, 0.3–3.6).

Figure 3B shows changes in LC stage distribution at diagnosis from 2005 to 2019. While the stage distribution in men remained unchanged, women experienced a stage shift. Distant-stage diagnoses, which constituted half of LC cases in women in 2005, fell to 39% by 2019. Localized-stage cases increased from 23% to 36%. By 2019, localized and regional-stage cases combined represented 61% of all new LC diagnoses, rising from 50% in 2005.

Figure 4 details trends in the age of LC diagnosis by stage from 2005 to 2019. In men, the mean age at diagnosis increased across all stages, reflecting population aging. In women, the mean age at distant-stage diagnosis also increased (APC, 0.5%; 95%CI, 0.5–0.6), but

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remained stable for localized-stage (APC, −0.1%; 95%CI, −0.2–0.0; p=0.2), with a slight decline by 2019.

Yearly analysis confirmed these patterns (Supplementary Figure 2). Diagnosis age in men increased across all stages, whereas in women, it decreased for localized stage and increased for distant-stage. Women's ages at regional-stage diagnosis developed a bimodal distribution, indicating convergence between localized and distant stages.

Improvements in 5-year survival rates outpaced reductions in mortality (Supplementary Figure 3). Comparing cases diagnosed between 2016–2020 and 1993–1995, 5-year survival more than tripled in women (from 15.8% to 50.3%) and more than doubled in men (from 11.6% to 30.6%). This improvement was consistent across all stages. Distant-stage survival more than doubled in both genders over the last decade. In women, localized-stage survival increased from 59.2% (2006–2010) to 87.6% (2016–2020), while regional-stage survival rose from 35.9% to 63.6%.

From 2010 to 2022, 157,566 men and 79,607 women underwent lung surgeries, of which 57,138 (36.3%) and 38,659 (48.6%) were LC surgeries, respectively. Figure 5 shows an increase in surgeries for both cancer and benign conditions, more pronounced in women. In men, lung surgeries increased from 9,111 to 15,083, and LC surgeries from 2,810 to 5,410. Age-standardized rates for lung and LC surgeries increased annually by 2.6% (from 15.8 to 19.4 per 100,000) and 1.9% (from 41.2 to 56.1 per 100,000), respectively. In women, lung surgeries increased from 3,565 to 8,595, and LC surgeries from 1,365 to 4,375. Age-standardized rates increased annually by 6.5% (from 16.4 to 31.8 per 100,000) and 7.1% (from 6.8 to 16.0 per 100,000), respectively. Additionally, limited resections constituted a significant proportion of all lung surgeries. In 2022, wedge resections and segmentectomies comprised 69% of lung surgeries in men and 59% in women (Supplementary Figure 4). Despite increasing surgeries, nonsurgical lung biopsy rates remained unchanged (Supplementary Figure 5). Rates of bronchoscopy (men; AAPC, −1.3%; 95%CI, −2.2–−0.4; women; AAPC, −0.9%; 95%CI, −1.7–−0.2) and mediastinoscopy (men; AAPC, −20.5%; 95%CI, −22.6–−18.2; women; AAPC, −17.6; 95%CI, −21.9–−13.0) decreased, while thoracoscopy rates slightly increased (men; AAPC, 1.3%; 95%CI, 0.1–3.1; women; AAPC, 2.3%; 95%CI, 0.7–3.9).

Discussion

This nationwide cohort study, analyzing Korea's national registry and health check-up records of 1.7 million adults revealed: 1) widespread self-initiated, non-reimbursed LDCT screening, 2) LC overdiagnosis in women, and 3) increasing lung surgeries bypassing nonsurgical biopsies. Indicators of LC overdiagnosis and overtreatment emerged in a largely non-smoking population engaging in unregulated screening. To our knowledge, this is the first study to quantify self-initiated, non-reimbursed LDCT screening prevalence and to reveal the discrepancies between recommended and actual screening practices, highlighting public enthusiasm for potentially unnecessary screening. Moreover, it first reports national-level changes in medical practices and potential overtreatment following

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excessive screening. Overall, this study advances our understanding of the consequences of over-screening, from overdiagnosis to increased healthcare utilization.

Recent studies underscored LC overdiagnosis in low-risk groups without quantifying unnecessary screenings and their repercussions.[10–12] A Taiwanese study involving 12 million women revealed a 6-fold rise in early-stage LC following intensive screening, without decreased advanced-stage diseases or mortality.[10] Corroborative patterns emerged in Korea[11] and China.[12] Asian screening trials showed comparable LC incidence in individuals with minimal smoking exposure (0.45–2.22%) and those with significant smoking history (0.86–1.36%), with most cases being indolent early-stage disease.[23,24,42] The Taiwan LC Screening in Never Smoker Trial reported a 2.6% detection rate (96.5% being stage 0 or I indolent adenocarcinoma), surpassing rates in high-risk populations, including the National Lung Screening Trial (NLST) (1.1%) and Netherlands-Leuvens Longkanker Screenings Onderzoek (NELSON) trial (0.9%).[5,7,8,25]

We extend this by showing that LC diagnosis in women is increasingly occurring at younger ages and earlier stages. Additionally, we reveal a high level of interest in guideline-discordant screening. Despite only 6% of women smoking and 0.04% qualifying for risk-based screening, 46% underwent non-reimbursed LDCT. Concurrently, lung surgery rates for both malignant and benign lesions increased, with the majority being limited resections, suggesting its growing use for biopsy purposes.[43] In 2022, lung and LC surgery rates were 31.8 and 16.0 per 100,000 women, while nonsurgical lung biopsy rates remained at 11.1 per 100,000. The steep rise in LC surgeries in women—despite lower LC incidence rates compared to men—indicates potential overtreatment. In 2019, the rate of LC surgery in women was 16.2 per 100,000, compared to localized and regional LC incidence rates of 13.7 and 9.3 per 100,000. This suggests that many regional-stage cases were also operable, aligning with the trend toward earlier-stage diagnoses.

Multiple factors contribute to the widespread uptake of screenings. Primarily, the high accessibility of LDCT—enhancing risk-based LC screening participation rate to 52.7%, exceeding Europe (7.3%)[44] and the U.S. (6.5%)[45]—inadvertently encourages screening in low-risk groups less likely to benefit. Second, extensive media coverage of survivor stories and the virtues of early detection perpetuates the “populatory paradox,” where increased screening causes more overdiagnosis, reinforcing perceptions that screening saves lives thus gaining popularity, while in reality exposing patients to potential harm.[22] Public messages easily emphasize screening-related benefits and the influence of risk factors (e.g., family history of LC, secondhand smoke, air pollution, and cooking fumes) but downplay harms, including overdiagnosis, overtreatment, complications from evaluating and managing screen-detected abnormalities, and increased healthcare expenditures.[46–48] Third, public enthusiasm toward cancer screening remains high, despite limited awareness of its potential harms. A U.S. study found that 87% of adults view cancer screening almost always positively, with 74% believing early detection saves lives “most” or “all of the time”.[49] Despite 38% experiencing false positives, 98% expressed satisfaction, and two-thirds wanted screening even if nothing could be done. A population-based survey in the UK found that while 54% participated in cancer screening, only 29.2% were aware of overdetection, and 0.2% had seen it mentioned in screening leaflets. Respondents considered 113–313

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overdetection cases per 1,000 screenings acceptable.[50] Lastly, our limited understanding of risk factors and the natural history of LC in never-smokers (LCINS) amplifies willingness to be screened, due to inherent discomfort with uncertainties.[51]

Korea's female LC "epidemic" appears more attributable to overdiagnosis than an actual rise in clinically significant cancers. Typically, shifts in cancer occurrence show parallel changes in incidence and mortality, influenced gradually by environmental exposures or genetic inheritance.[21] However, rapidly increasing early-stage LC incidence without corresponding mortality changes suggests external influences from medical practices.[21] Effective screening reduces mortality by detecting potentially symptomatic cancers earlier, increasing early-stage incidence while decreasing late-stage incidence.[22] Overdiagnosis, however, inflates early-stage incidence without reducing late-stage incidence and mortality,[28] with its extent estimated by the excess of early-stage diagnoses over late-stage reductions.[36] Although the slight decrease in distant-stage LC may reflect screening's intended benefits, much largely and abruptly increased localized-stage cases in low-risk women—even in the era of screening and improved staging practices for advanced disease—challenges the assumption that certain risk factors selectively cause localized-stage LC in younger women.[21,22] Furthermore, the growing LC prevalence, despite stable smoking rates and risk exposures, cannot be explained by new etiologic factors or increased exposure to known risks.[52] Improved survival across all stages, outpacing mortality reductions—the most powerful misleading feedback from overdiagnosis—likely reflects treatment advances rather than screening benefits.[21,22,53] Altogether, overdiagnosis with treatment advancements beyond stable LC risk likely explains this "epidemic". This resonates with breast and thyroid cancer epidemiology following extensive screening.[36,54]

LC overdiagnosis is pronounced in women with low smoking rates, due to the unique biology of LC in this demographic. While the NLST reported a 3% overdiagnosis rate in high-risk populations, low-risk populations exhibit different dynamics.[55] LCINS, mostly early-diagnosed adenocarcinomas, represent a distinct, indolent disease entity with actionable driver mutations conducive to better survival.[56] Over 95% of screen-detected LCINS appear as SSNs, of which 90% are adenocarcinoma spectrum lesions, from precursor glandular lesions to invasive adenocarcinomas.[18] Early adenocarcinoma spectrum lesions show minimal growth over extended periods, maintaining their early-stage status over time.[57] Analyzing 2898 SSNs showed only 22% growing over 2 years,[15] with growth thresholds—≥2mm increase in total SSN diameter, ≥2mm increase in solid portion, or new solid portion development—overestimating progression without affecting clinical outcomes.[58] SSNs stable for 2–5 years showed 5% growth rates and 2–13% beyond 5 years,[16,17] with stage shifts occurring at 9 years for part-solid nodules and 12 years for pure ground-glass nodules.[57] Increased screening intensity directly raises the risk of diagnosing slow-growing cancers, leading widespread LDCT to uncover many indolent early-stage LCINS.[27,28] In China, indiscriminate screening raised early-stage LCINS in younger women, with 93% slowly progressing and 7% even regressing.[59] Despite increased screening in men, overdiagnosis was not evident. Several factors may explain this. First, given the high smoking prevalence among men, screening may have identified many smoking-related, clinically significant LCs, rather than inflating overdiagnosis. Second, LC in men often exhibits aggressive behavior (particularly when associated with smoking, but

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even when controlling for smoking history) making it less susceptible to overdiagnosis.[60–63] Overdiagnosis occurs when screening detects slow-growing or non-progressive cancers, whereas fast-growing cancers frequently cause symptoms or mortality before detection.[22] Third, declining smoking rates may have reduced clinically significant LCs, yielding a less proportional increase in detection rates despite more screening. Lastly, a higher rate of clinically diagnosed versus screen-detected LCs may have influenced observed trends, warranting future studies to evaluate changes in these proportions.

Increasing surgery rates, not matched by biopsy increases, reflect Korea's current approach to managing screen-detected SSNs. Variable diagnostic yield of nonsurgical biopsies for small-sized SSNs, the requirement for entire lesion for accurate histopathologic diagnosis of adenocarcinoma spectrum lesions, and difficulties in predicting invasiveness and progression, often cause patient and physician anxiety, prompting a preference for direct surgical resection.[64–66] Divergent guidelines on pre-operative pathological diagnosis, combined with advanced surgical capabilities and excellent post-operative outcomes, have contributed to a trend to resect screen-detected SSNs without histologic confirmation.[67–69] Korea's single-center study showed 92.7% of SSNs being resected without histologic confirmation.[69] Similarly, in the Netherlands, over one-third of suspected early LC patients underwent surgery without histologic diagnosis.[67] This raises concerns about overtreatment since a compilation of studies shows up to 22% of surgically removed lesions are benign,[70] and even malignant lesions exhibit indolent behavior. Compounding this are non-negligible perioperative complications, with operative mortality rates of up to 1.2% and overall complication rates of 24%, even for minimally invasive thoracic surgeries.[71–75] Equally concerning is that patients may experience sustained pain, impaired lung function, increased healthcare costs, reduced functional capacity, and diminished quality of life for years following surgery.[76–78]

This study underscores the need for evidence-based LC screening and management strategies, supported by consensus among clinicians, policymakers, and the public. First, screening should prioritize reducing mortality over increasing detection rates, by targeting individuals at risk of lethal LC who would benefit from early intervention. Second, randomized controlled trials (RCTs) should assess the mortality benefits and potential overdiagnosis of screening populations outside current eligibility standards, such as those with minimal smoking exposure or a family history of LC. Third, lung nodule growth assessment and risk prediction models should be used in clinical practice to ensure invasive procedures are reserved for cases where growth affects prognosis. Fourth, clinically relevant growth thresholds should be established to guide optimal interventions for SSNs, as the current threshold may lack clinical significance. Fifth, long-term trajectories of screen-detected SSNs should be investigated in future studies. Sixth, given the non-negligible occurrence of aggressive LC in low-risk populations, efforts should focus on understanding the heterogeneity and diverse progression pattern of LC and devising tailored risk stratification and screening strategies, contingent upon RCTs showing a mortality reduction. Seventh, governmental and organizational efforts are needed to tailor public screening strategies. In Korea, many local governments and companies started funding LDCT for school cafeteria workers and general employees. In Taiwan and China, LDCT is regularly offered to teachers, firefighters, women from lower socioeconomic backgrounds, and in-

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hospital employees. Yet, the cost-effectiveness and safety, require thorough evaluation. Eighth, clinicians should discuss the benefits and risks of LC screening based on personalized risk assessments, engaging patients in shared decision-making. Finally, while acknowledging overdiagnosis, the benefits of early detection in some cases should not be overlooked. Certain clinically meaningful LC cases may be increasing, and some populations traditionally considered low-risk for developing aggressive LC may indeed benefit from screening. Balanced approaches recognizing their coexistence are crucial. Korea's experience provides cautionary lessons globally. Although Korea faced pronounced overdiagnosis during the thyroid cancer "epidemic", substantially decreased thyroidectomies post-2014 demonstrate positive changes driven by increased recognition.[54,79] This indicates that the extent of LC overdiagnosis and overmedicalization can also be reversed through increased awareness and strategic interventions.

This study has several limitations. First, non-reimbursed screening rates were evaluated in a 1.7 million cohort, not the entire Korean population. However, the cohort's size and diversity likely reflect nationwide trends. Second, data acquisition periods varied across sources, but all available data was compiled and analyzed to accurately capture contemporary trends. Third, LC surgeries were defined as those involving mediastinal lymph node dissection with lung resection, potentially excluding cases where N2 dissection was not performed (e.g., curative lobectomy without N2 sampling). Fourth, unaccounted confounding factors may have affected LC epidemiology and healthcare utilization. Despite complexities in interpreting 5-year survival and age at diagnosis, these were included to provide a comprehensive understanding.

In conclusion, this study identified a widespread practice of self-initiated, non-reimbursed LDCT screening in a predominantly non-smoking population, associated with LC overdiagnosis and increased lung surgeries bypassing nonsurgical biopsies, particularly in women. This underscores the potential impact of screening conducted outside of guideline-directed care, which may pose harm to low-risk individuals for developing lethal LC and lead to increased healthcare utilization with limited benefit. RCTs comparing LDCT versus usual care for individuals with minimal smoking exposure, with or without a family history of LC, are needed to evaluate mortality reduction before implementation beyond established guidelines. Adjusting LC screening and management strategies is crucial to balance early detection benefits with the harms of overdiagnosis and overtreatment.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Sharing Statement

All study data will be available immediately after publication to researchers who submit a methodologically sound proposal that meets the aims of the approved proposal. Proposals should be directed to the corresponding author (YSP) at mdyspark@snu.ac.kr.

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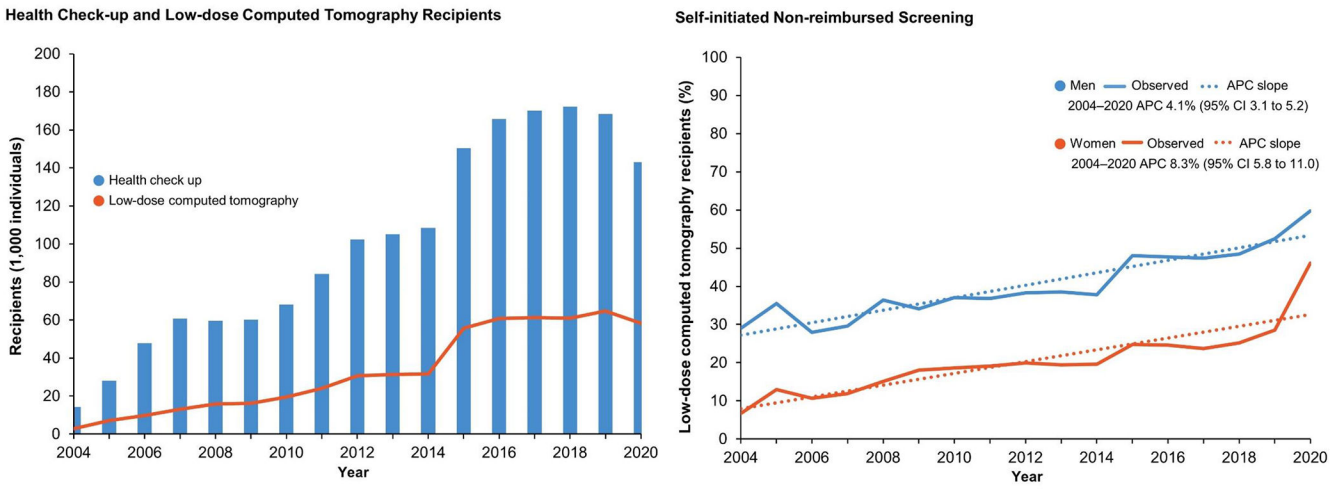


Figure 1. Trends in health check-ups and non-reimbursed low-dose computed tomography screening, 2004–2020.

Annual trends in health check-up attendance and non-reimbursed low-dose computed tomography (LDCT) screening, and the proportion of individuals opting for LDCT during check-ups. APC indicates annual percentage change.

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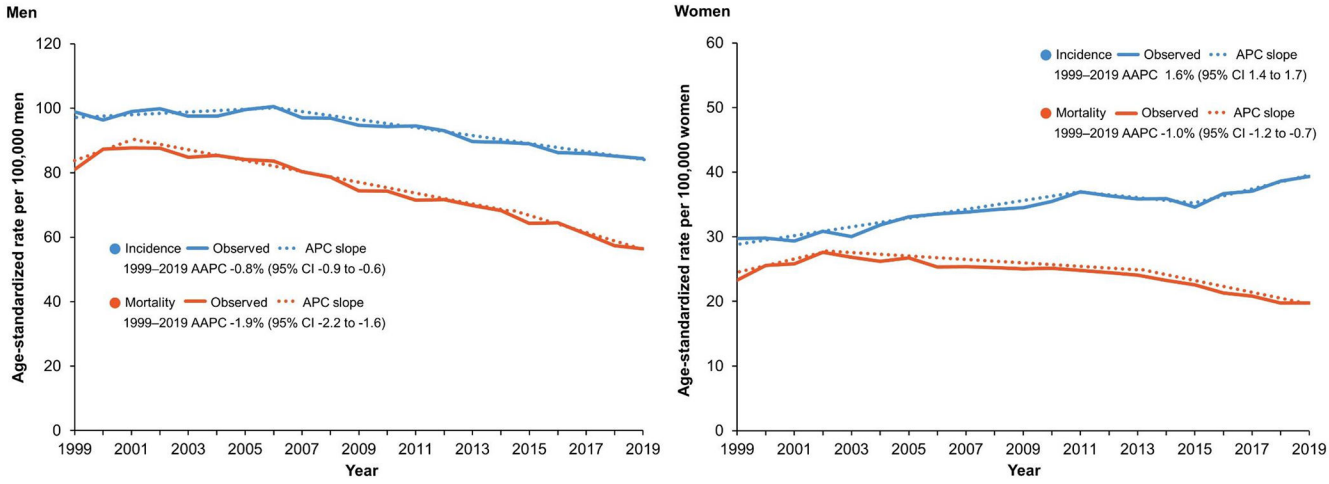


Figure 2. Lung cancer incidence and mortality rate in Korean men and women, 1999–2019.

Trends in age-standardized lung cancer incidence and mortality rates in Korea from 1999 to 2019. APC indicates annual percentage change and AAPC indicates average annual percentage change. Data was obtained from Statistics Korea.

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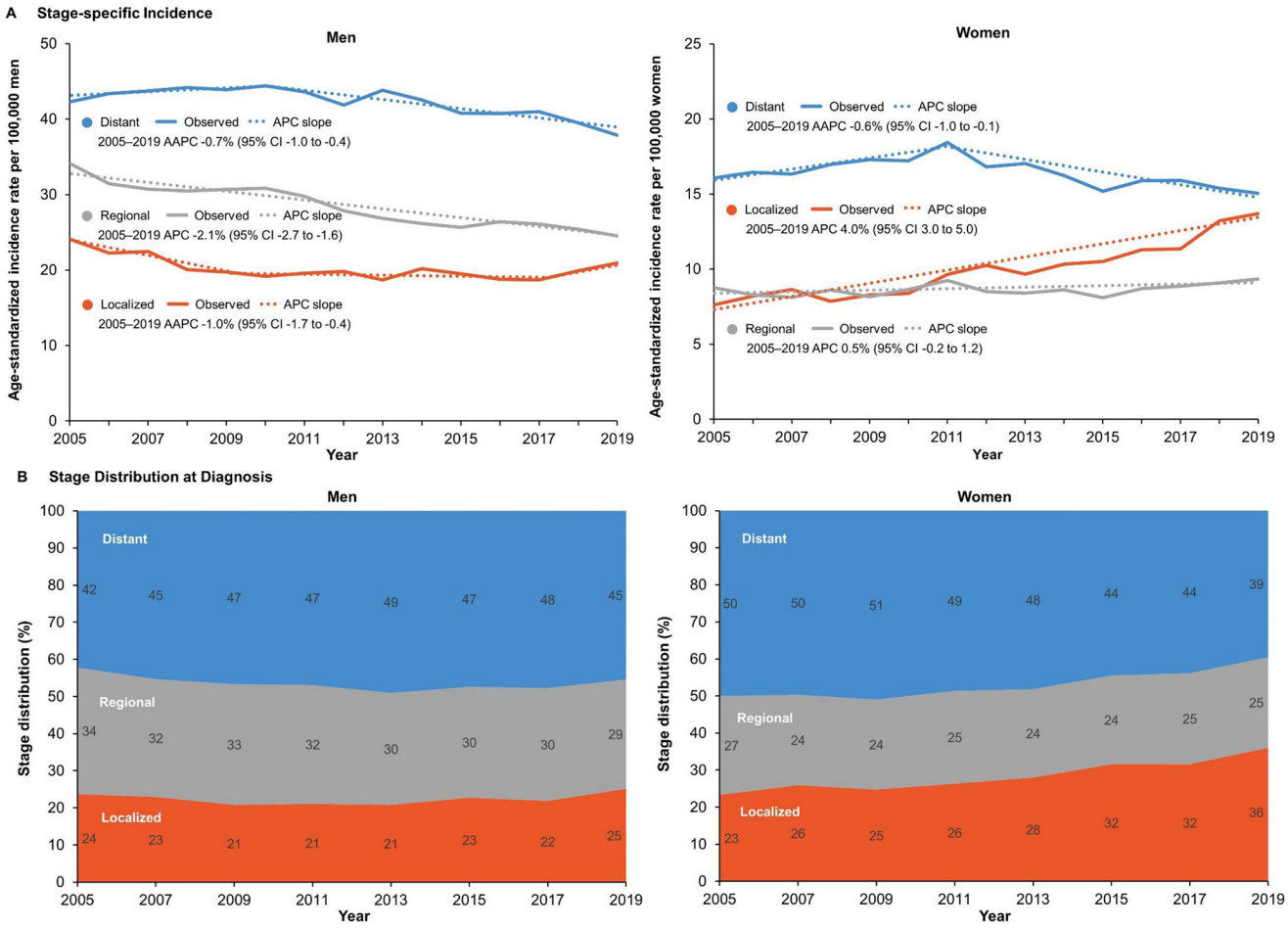


Figure 3. Lung cancer stage-specific incidence rate and stage distribution in Korean men and women, 2005–2019.

Trends in age-standardized incidence rates by stage (Figure 3A) and stage distribution at diagnosis (Figure 3B) for lung cancer in Korea from 2005 to 2019. APC indicates annual percentage change and AAPC indicates average annual percentage change. Data was sourced from the Korea National Cancer Incidence Database (KNCI DB).

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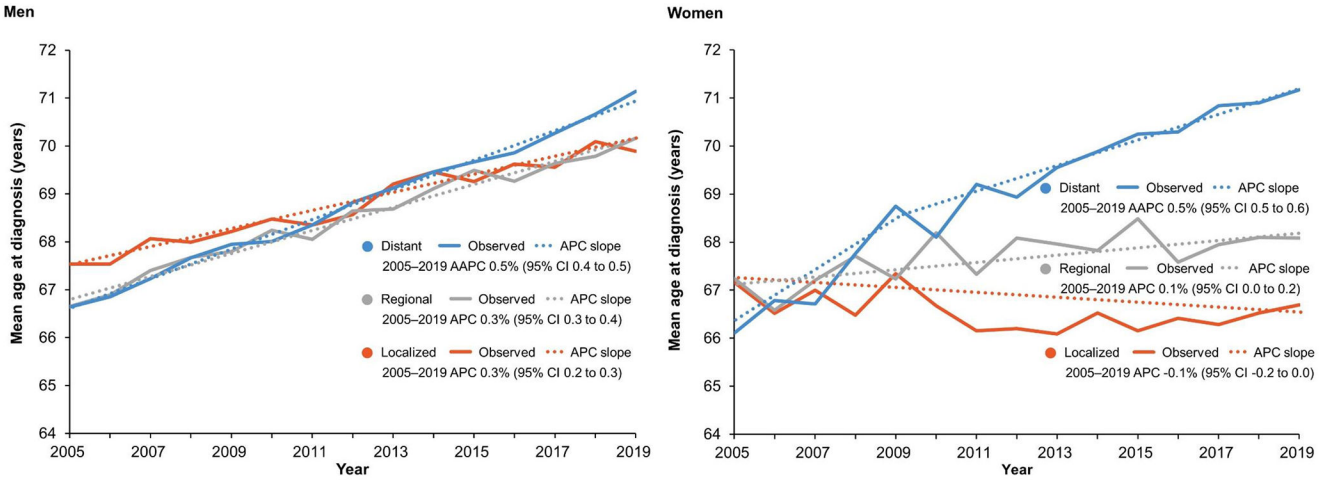


Figure 4. Age at lung cancer diagnosis by stage in Korean men and women, 2005–2019.

Trends in age at lung cancer diagnosis by stage in Korea from 2005 to 2019. APC indicates annual percentage change and AAPC indicates average annual percentage change. Data were sourced from the Korea National Cancer Incidence Database (KNCI DB).

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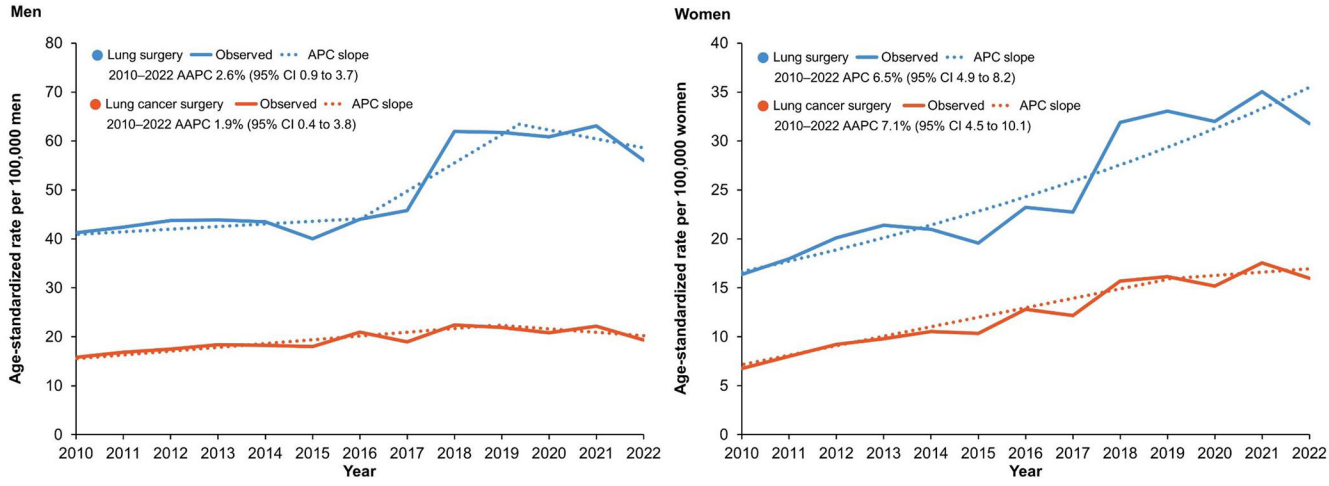


Figure 5. Lung surgery rate in Korean men and women, 2010–2022.

Trends in age-standardized rates of lung and lung cancer surgery in Korea from 2010 to 2022. APC indicates annual percentage change and AAPC indicates average annual percentage change. Data was obtained from the Korean Health Insurance Review and Assessment (HIRA) database.